

Usability Testing: ByteBreakers

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Study Overview

- **11 scenarios were tested** across
 - signup, 2FA, posting, messaging, and archiving.
- Both researcher and organization roles.
- Observe how easily they navigate and interact with ByteBreakers.
- Screen and audio are recorded (with consent) for internal, confidential analysis.

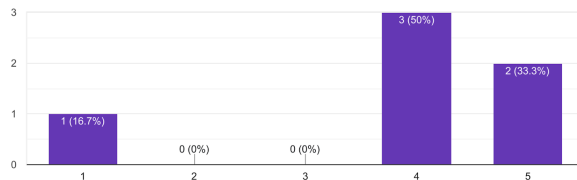
Objectives

- Evaluating task flow efficiency, perceived security, user pain points with feedback and ratings, navigation, security trust, and overall UI/UX comfort to generate actionable design recommendations.

Methodology & Analysis

- 6 One-hour sessions.
- Screen/audio recording, task tracking.
- Post-test interviews & Combined task metrics.
- Data was then transcribed, and qualitative coding was done.
- Qualitative coding and analysis were performed.

On a scale of 1-5, how concerned are you about your digital privacy and security?
6 responses

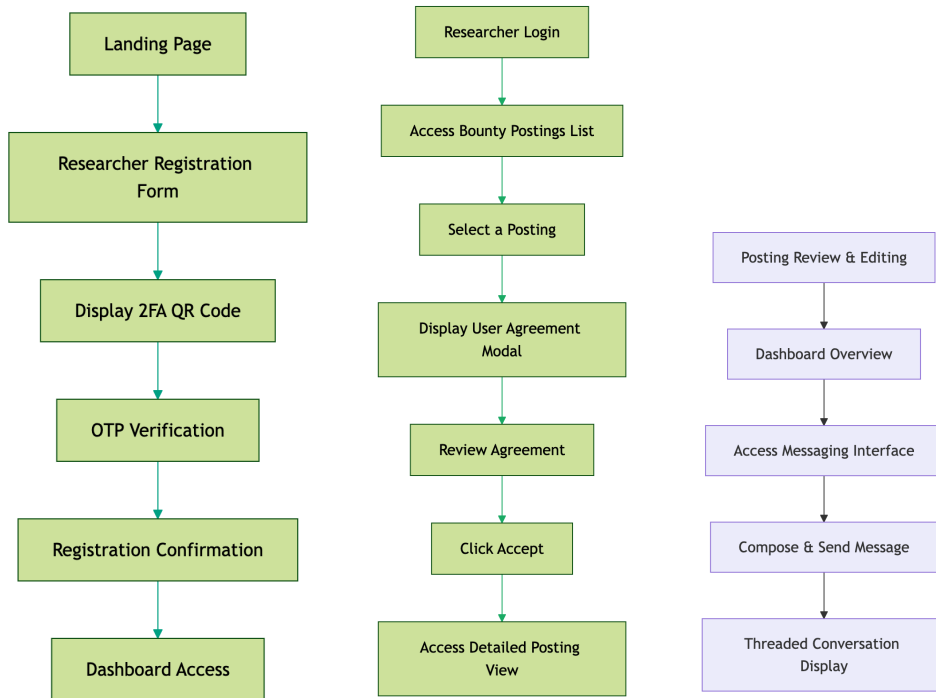


Participant Demographics

- Ages 23–26.
- Grad students (mixed backgrounds).
- Broader OS exposure.
- Those who were Very or Extremely familiar with web applications reported higher privacy concern levels, suggesting a link between awareness and caution.

User Flow Charts:

Sample Flow Diagrams for Tasks:



Test Scenarios & Tasks

- New User Registration: Assess clarity and security of researcher sign-up and 2FA flow.
- User Browsing & Agreement Acceptance: Evaluate ease of browsing and agreement interaction.
- Org Login with Researcher Credentials: Test feedback clarity on incorrect role-based access.
- Organization Registration & Onboarding: Examine registration smoothness and onboarding clarity.
- Creating Bounty Postings: Measure the ease and usability of the markdown editor for new posts.
- Editing Bounty Postings: Check edit discoverability, interface clarity, and confirmation cues.
- Secure Messaging: Validate intuitiveness and perceived security of message exchange.
- Org Signup without Sample Data: Gauge first impressions and clarity without onboarding content.
- Archiving & Submission Workflow: Observe editability and system feedback post-archival.
- Attempting Unarchive Operation: Test clarity of restrictions and system messaging.
- Change Password & Relogin: Verify password update process and secure login behavior.

Post-Test Interview Questions

- **General Usability & Navigation:**
Evaluate ease of use, task clarity, and overall flow throughout the platform.
- **Security Perceptions:**
Gauge user trust, understanding, and comfort with security features like 2FA and messaging.
- **Error Handling & Feedback:**
Assess clarity, helpfulness, and actionability of error messages and system feedback.
- **UI/UX & Visual Design:**
Explore impressions of visual consistency, design aesthetics, and interface usability.
- **Detailed Functionality Feedback:**
Identify unnecessary steps, information clarity, and feedback effectiveness during specific tasks.

Participant Ratings						
 P#	Overall Experience (1-10)	Security Perception (1-10)	Visual Design (1-10)	Navigation Ease (1-10)	Onboarding Experience (1-10)	Would you Recommend
P1	3.0	5.0	3	5	5	Maybe (if improved)
P2	4.0	4.5	4	6	6	Yes (after fixes)
P3	7.5	7.0	7	8	7	Yes
P4	9.0	9.0	8	8	9	Yes (after fixes)
P5	7.5	6.0	8.5	6.0	5.0	Yes (after fixes)
P6	9	9.0	8.0	9.0	9.0	Yes
Average Score (out of 10)	6.67	6.75	6.42	7.0	6.83	

Testing results:

- **Overall User Experience:** The concept was understood, but the user flow felt inconsistent and occasionally disjointed.
- **Onboarding & Authentication:** Users understood the concept, but onboarding—especially for organizations—lacked direction. Improve sign-up flows with role-specific guidance, visual cues, and clear, step-by-step 2FA setup instructions.
- **Navigation & Information Architecture:** Navigation felt smoother for researchers but disjointed for organizations. Add consistent sidebar or top menus, use tabs/panels to organize content, and include tooltips to improve orientation.
- **Task Flows, Forms & Posting:** Tasks like posting and editing were completed but often involved friction due to unclear tags or small buttons. Clarify tag input, improve form spacing, and allow preview and undo actions for posts.
- **Security & Access Control:** 2FA was valued but caused confusion after QR scanning; users also flagged weak password flows. Enhance security flows with inline guidance, block incorrect role-based access, and show clear validation feedback.
- **Feedback & Error Messaging:** System feedback helped at times but was too subtle or brief in others. Use clearer, persistent messaging in visible areas, and ensure all feedback is actionable and easy to understand.
- **Visual Design & UI Consistency:** Clean visuals were appreciated, but minimalism hurt clarity in some areas. Improve contrast, font readability, and visual hierarchy, while ensuring consistent design across roles and screens.
- **Functionality Enhancements:** Users missed sample data and contextual cues after sign-up. Add dummy data, enable profile previews in submissions, implement “Forgot Password” with OTP, and support in-app feedback/reporting tools.

Example:

Security Perception

Security was a recurring theme in the interviews. Most participants felt the inclusion of two-factor authentication added credibility. However, a few technical users noticed flaws like being able to change a password without proper

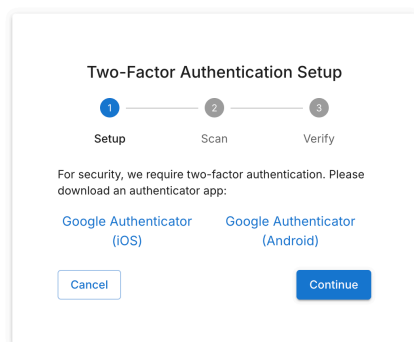
validation or using the same login credentials across different roles. These loopholes made the platform feel less secure to them, despite the presence of other security features.

Login page refinements:

For your security, 2FA with Google Authenticator is required when signing up. Scan the QR code and enter the security code below

ByteBreakers

Bug bounty platform for you to show off your hacking skills



Two-Factor Authentication Setup

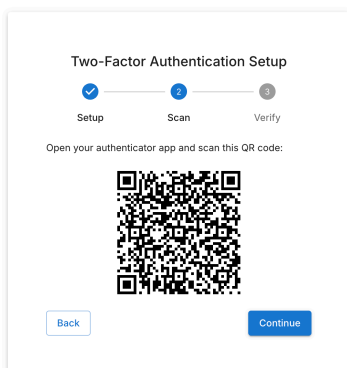
1 Setup 2 Scan 3 Verify

For security, we require two-factor authentication. Please download an authenticator app:

[Google Authenticator \(iOS\)](#) [Google Authenticator \(Android\)](#)

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Two-Factor Authentication Setup

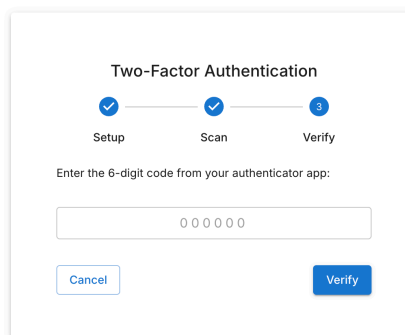
1 Setup 2 Scan 3 Verify

Open your authenticator app and scan this QR code:



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Two-Factor Authentication

1 Setup 2 Scan 3 Verify

Enter the 6-digit code from your authenticator app:

Chat page refinements:

Participants found the chat feature either hard to access or cluttered. One felt the layout was overwhelming, and another couldn't locate it on the organization side. The lack of sample messages made it unclear how the chat worked, highlighting the need for a simpler design and test data.

Comments (1)

 Seemal Tausif

This is atest message

<> Write

 Preview

Type your message here... (Markdown supported)

****Bold****

Italic

``Code``

[Link](url)

Send 

Participant Insights

 P#	Common Pain Points	Suggested Improvements
P1	Confusing labels, missing confirmations, weak password recovery, and cluttered submission layout.	Improve labels, add 2FA-based password recovery, reorganize navigation, and show submission previews with profile info.
P2	Poor onboarding clarity, excessive white space, insecure org access, and vague error feedback. Insecure access to the organization dashboard.	Add sidebar navigation, clarify onboarding and color contrast, restrict org access by role, and enforce password rules.
P3	QR setup was unclear, the dashboard felt cluttered, the design lacked engagement, and the chat felt overwhelming.	Add color/animations, clarify onboarding, reposition dashboard items, and simplify the chat experience.
P4	QR instructions were confusing, tag input was unclear, feedback was too quick, and navigation lacked structure.	Clarify 2FA instructions, improve tag input and feedback visibility, add structured navigation, and separate post states.
P5	Tag entry was inconsistent, the org dashboard felt empty, and archive actions lacked clarity.	Add tag input hints, enhance the org dashboard layout, and clarify/archive feedback with undo.
P6	Mostly smooth flow, but minor UI polish was needed, and manual org verification was unclear.	Polish UI across roles and separate approved vs. archived posts for clarity.

Significant improvements were made in onboarding clarity, task flow guidance, and visual cues, leading to a smoother and more intuitive experience. Further enhancements should focus on refining error feedback, adding onboarding data, and unifying layout structure across roles.

Area	Old (P1–P4)	New (P5–P6)
Onboarding	Unclear start, confusing flow	Better for researchers, still rough for orgs
2FA	QR scan unclear, no next step	Clearer steps, minor OTP confusion
Navigation	No menus, hard to orient	Sidebar/menu helped, but not perfect
Visual Design	Bland, low contrast	Better colors, fonts, clearer layout
Forms & Tags	Tags confusing, buttons unclear	Tags still confusing, needs input hints
Error Feedback	Too fast, not visible	Better timing, color-coded, readable
Security	Role issues, weak password flow	Stronger 2FA, but some validation gaps

Usability Scores: Old vs New Participants

This chart compares the average usability ratings of early participants (Old Avg) with (New Avg) across five key categories. Every metric shows improvement- particularly in Visual Design, Security, and Overall Experience- indicating that recent updates to the Bytebreakers platform significantly enhanced the user experience.

Category	Old Avg (P1–P4)	New Avg (P5–P6)	Change
Overall Experience	5.88	8.25	+2.37
Security Perception	6.38	7.5	+1.12
Visual Design	5.5	8.25	+2.75
Navigation Ease	6.75	7.5	+0.75
Onboarding Experience	6.75	7.0	+0.25

Future Work: Improvements will include clearer post-creation guidance for the organization, the addition of dummy chat data, enhanced visibility for profile/chat/logout access, undo or confirmation options for critical actions, clearer password feedback, and better usability support for less tech-savvy users.